

Network Connections

02



Recall:

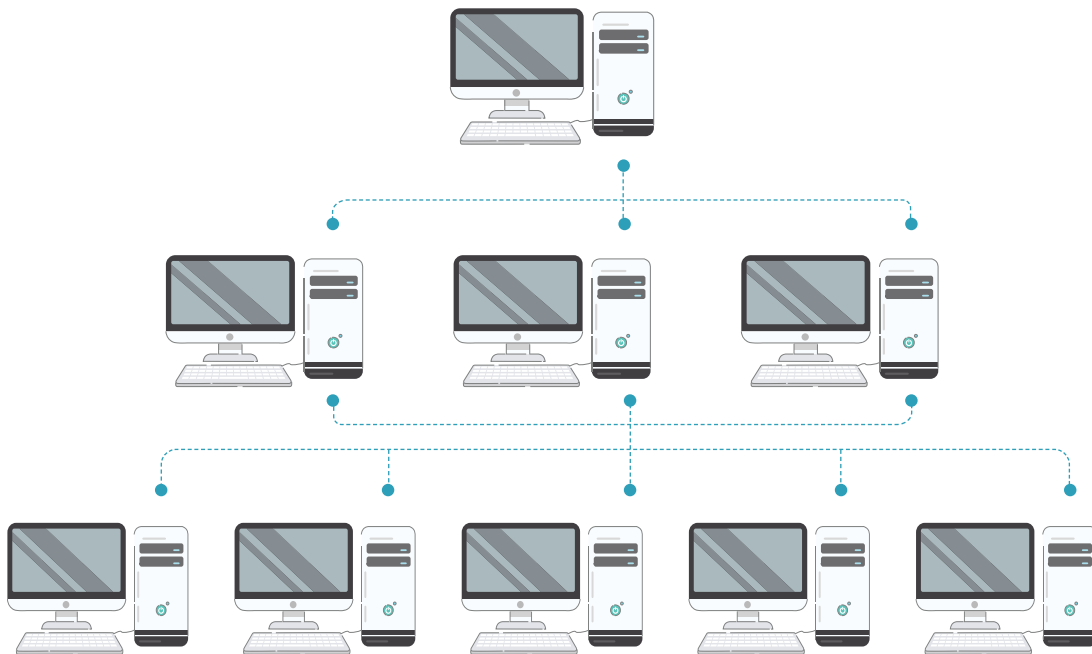
Internet is a global network of computers connected to each other. Computers communicate with each other through the internet. You can look for information about anything on the internet.

2. Network Connections

The internet is an important communication tool. We can use it to connect to anyone anywhere in the world. A computer network is the medium through which we connect to the internet.

What is a Computer Network?

A group of inter-connected computers is called a **computer network**.



Have you ever wondered how a computer is connected to the internet?
Different types of connections are used to connect a computer to the internet.

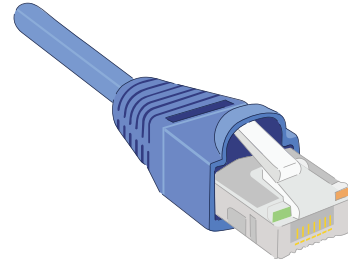
Bytes

The first working internet was introduced in the year 1960.

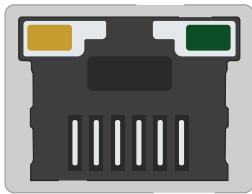
1. Wired Connections

1. Wired Connections: A wired connection makes use of wires or cables to connect to the internet.

a. Network Cables: These are wires attached to the computer to help it connect to the internet. These wires are known as **Ethernet Cables**. Ethernet Cables transport data from the computer to the internet, and then back to the computer.



Ethernet Cable

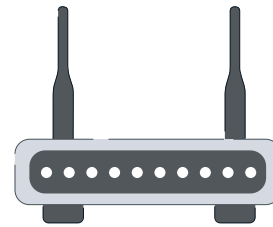


Ethernet Port

b. Ports: A port is a point (opening) in which the cable is connected. The ethernet cable is connected to a computer's ethernet port.

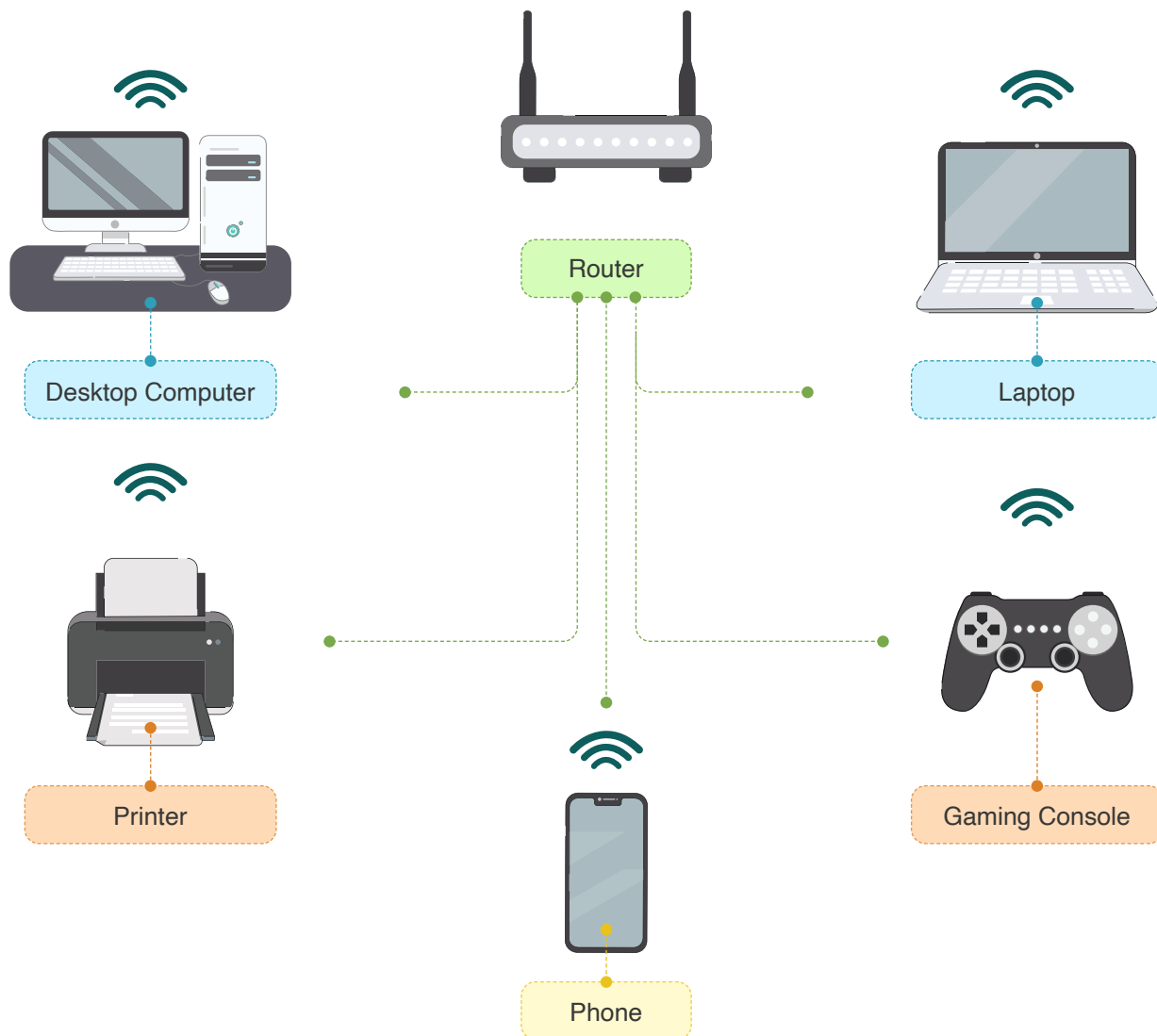
2. Network Connections

c. Router: A router is a device that can connect multiple devices (computers, mobile phones, tablets) in a house to the internet. The router sends internet signals to each device separately. Multiple devices can connect to the internet with just one router.



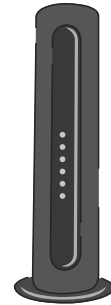
Router

How a router works



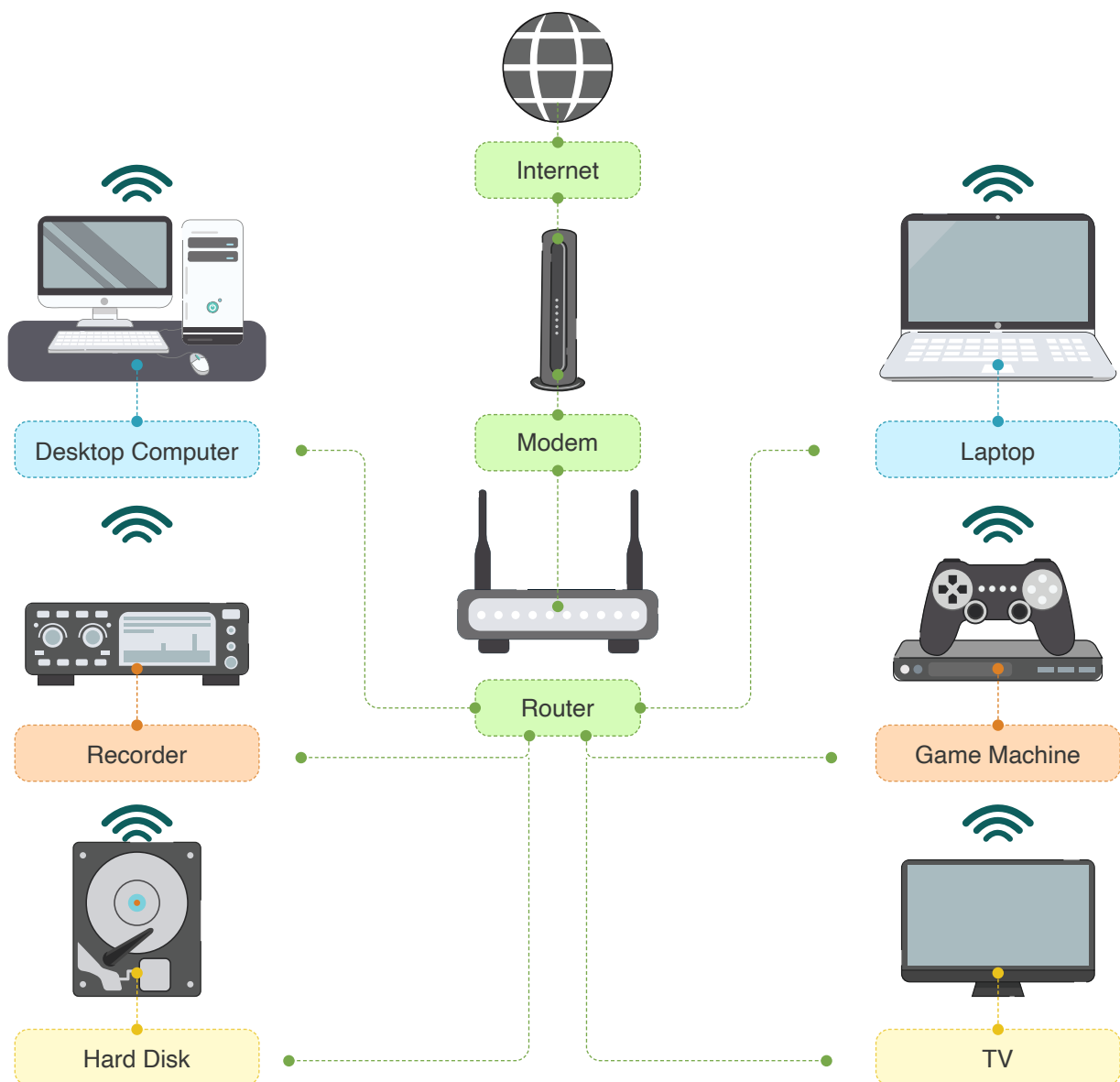
2. Network Connections

d. Modem: A modem connects a computer to the internet. It translates data sent from the internet into a language understood by a computer, and vice versa. If only one wired device is connected to the internet, the ethernet cable can be directly connected to the modem. A modem is an essential connection device.



Modem

How a modem works



Exercise

Fill in the blanks

- 1 A group of interconnected computers is called a _____.
- 2 The wire attached to a computer that connects it to the internet is called _____.

State whether True or False

- 3 A modem is not important to connect to the internet.

 _____

- 4 The router sends signals to each device separately.

 _____

- 5 We can only use wires to connect to the internet.

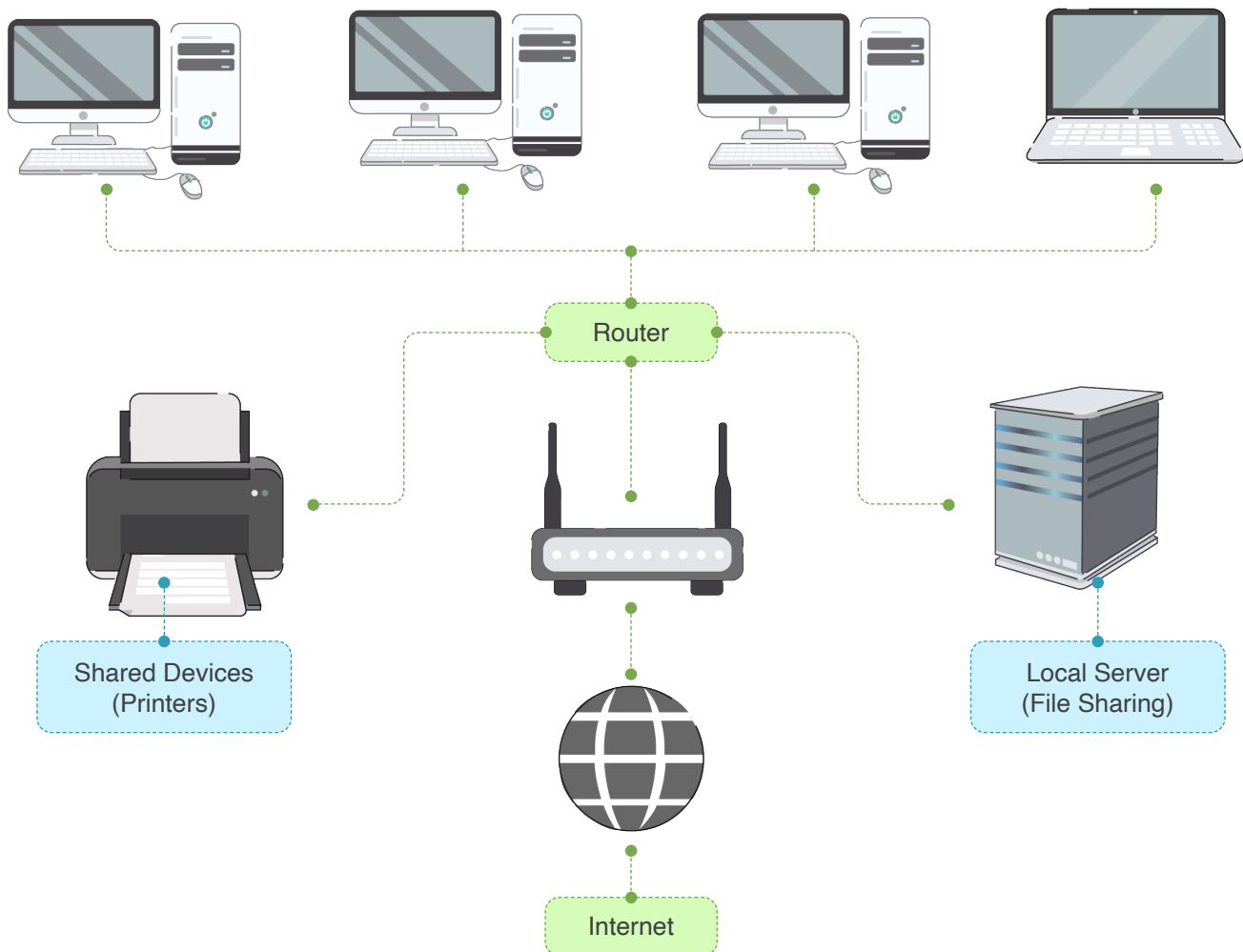
 _____

Note: Nowadays, we usually have more than one device that connects to the internet. Most places use a modem with router features.

2. Network Connections

e. LAN: Local Area Network (LAN) is a network of interconnected computers. A LAN network is limited to a small area, such a single room. The computers share information with each other.

For example: computers in school labs, office buildings, etc. are part of a LAN network.

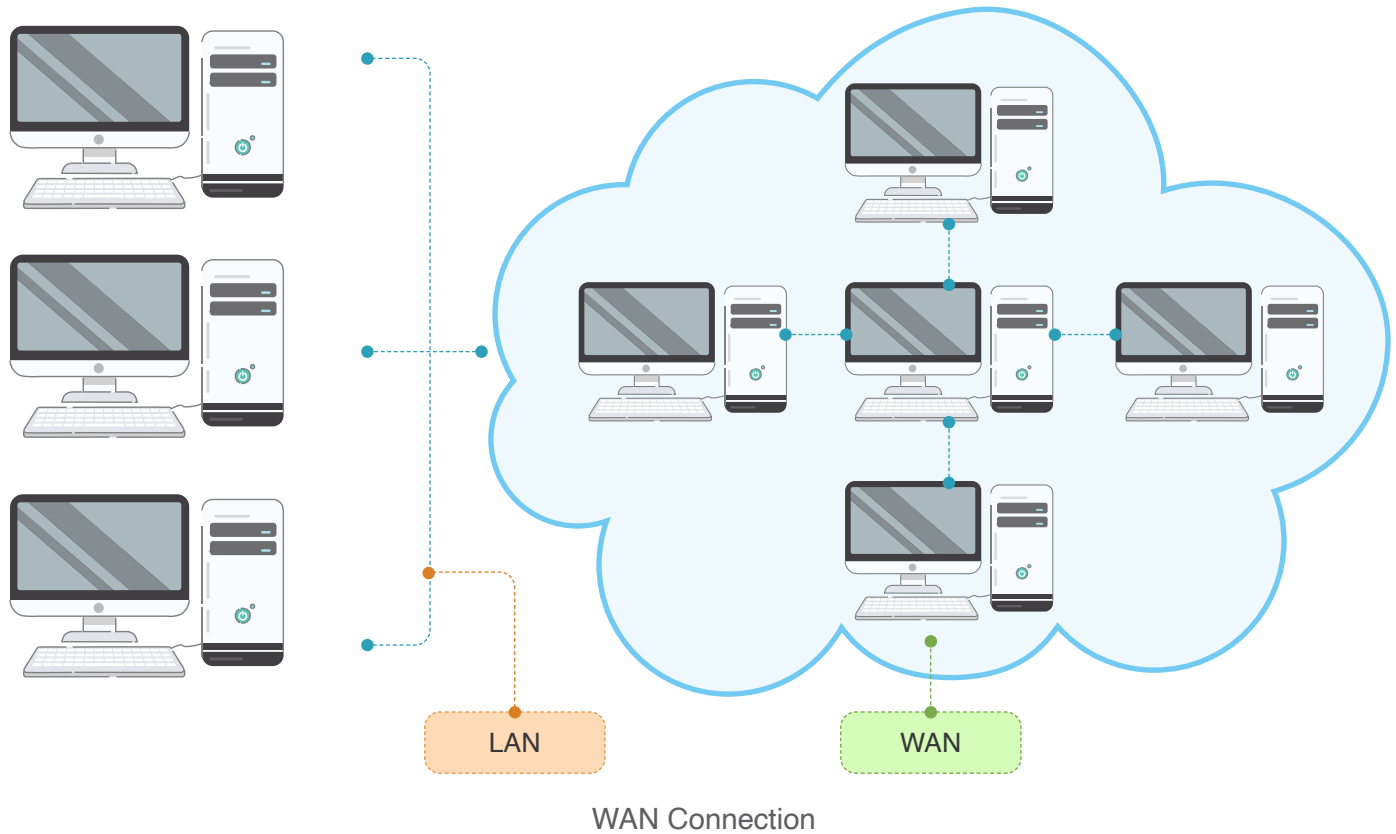


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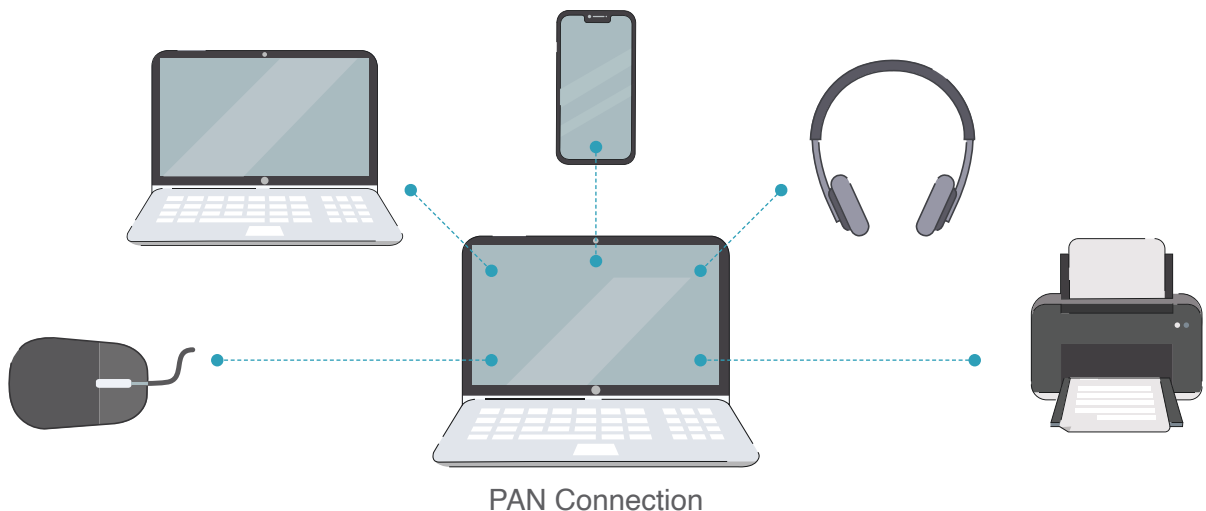
A server is a data storage destination. It acts as a common source of information for a group of computers.

2. Network Connections

e. WAN: Wide Area Network (WAN) is a network that connects many Local Area Networks (LAN). It covers large areas like a city or a country. The network connection between computers all over the world is a WAN.



f. PAN: Personal Area Network (PAN) is a network between devices that are used by a single person. **For example:** Headphones connected to a computer or phone, a printer connected to a computer, etc.



Exercise

Fill in the blanks

- 6 _____ is a network that is in a small office, computer lab or school.
- 7 _____ is a network that covers large cities and connects many LANs.
- 8 _____ is a network of devices that is used by a single person.
- 9 A _____ connects a computer to the Internet.
- 10 A _____ is used to connect multiple devices to the internet.

2. Wireless Connections

2. Wireless Connections: A connection which does not need any wires or cables to connect a device to the internet is called wireless connection.



Wi-Fi logo

2. Network Connections

Connect a computer to the internet through Wi-Fi

- 1 Click on the globe (network icon) on the taskbar.

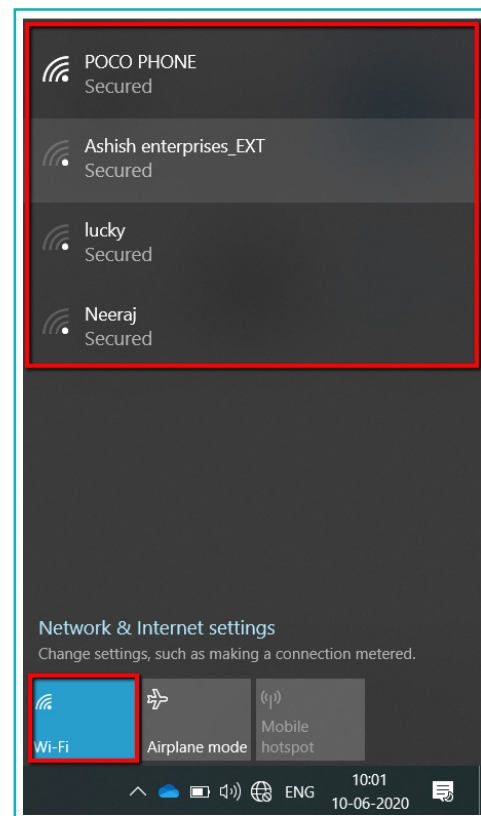


- 2 Check the list of available networks.

- 3 Select a network to connect to.

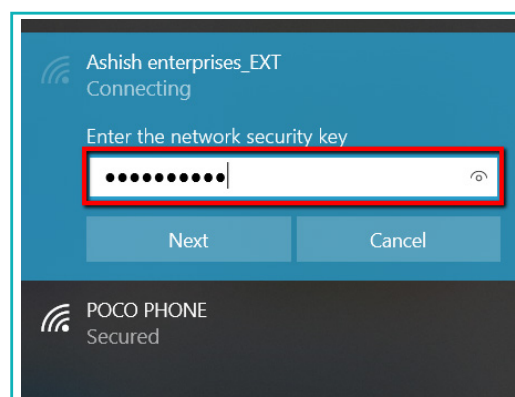
- 4 Enter the password for that network.

Note: People use passwords to protect their internet connection so that nobody else can log in and use it.



- 5 Click the 'Connect' button.

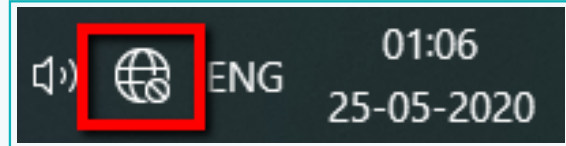
- 6 The computer should now be connected to the Wi-Fi network.



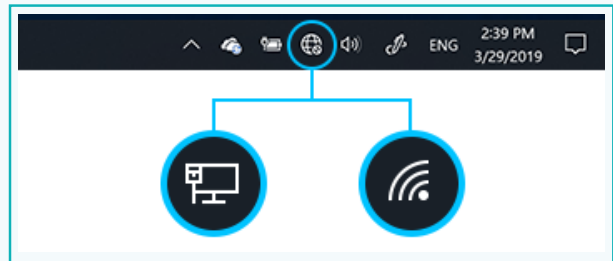
Activity 1

Let's look at how to check a device's internet connection.

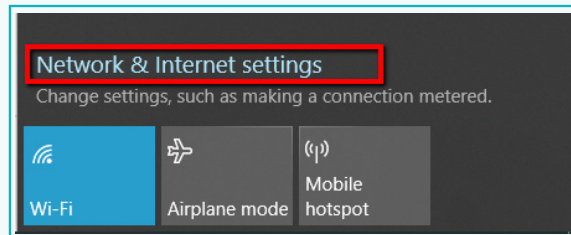
- 1 Find the network icon on the taskbar.



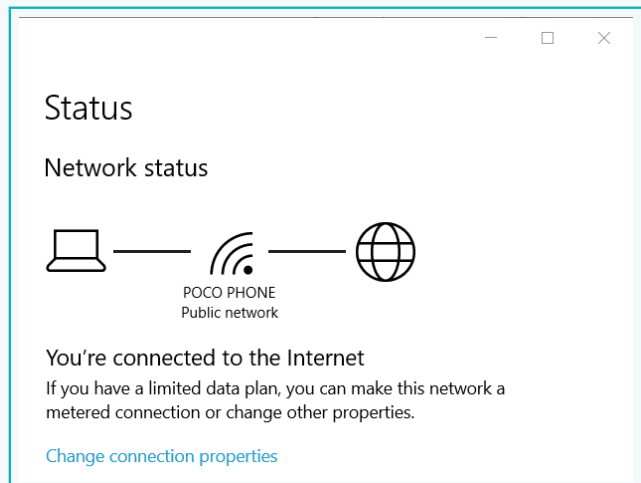
- 2 If the taskbar displays an icon for an ethernet or Wi-Fi Connection, the device is connected to the internet.
If not, the device is not connected.



- 3 To check the connectivity, click on 'Network and Internet settings'.



- 4 This will show the status of the device's internet connection.



Healthy Practices:

- Only use Wi-Fi networks which you recognise like the connection at school or home.
- Always use a password to keep your Wi-Fi connection protected.
- Turn off the Wi-Fi when not in use.

a. Bluetooth: Bluetooth is a technology which is used between devices to share data wirelessly over short distances. Bluetooth can be used to share documents, audio tracks, and videos. We commonly use a mobile phone, computer, headphones, speaker, keyboard, etc. to connect to via Bluetooth.



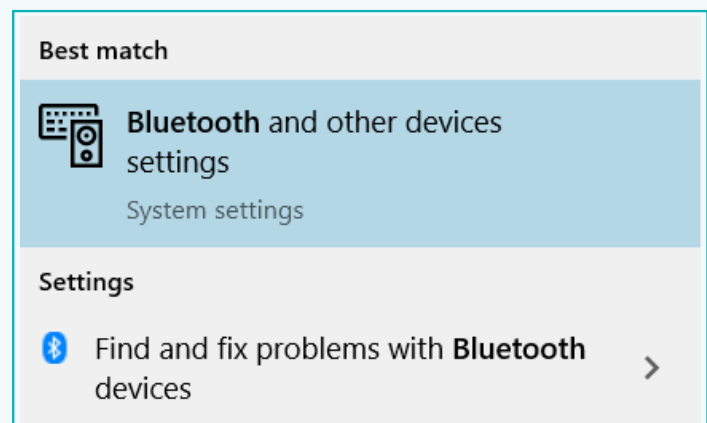
Bluetooth Icon

Activity 2

A Bluetooth connection eliminates the need for wires. Let's look at how to connect a device to a computer with Bluetooth.

1 Search for Bluetooth in the Search Bar.

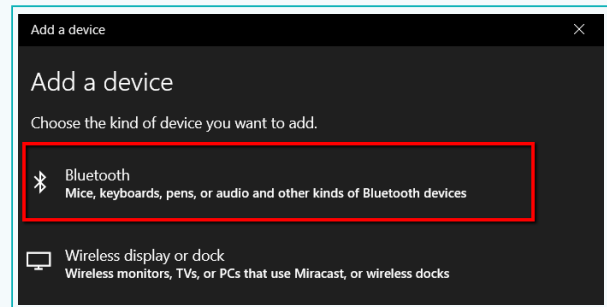
2 Click on 'Bluetooth and other device settings'.



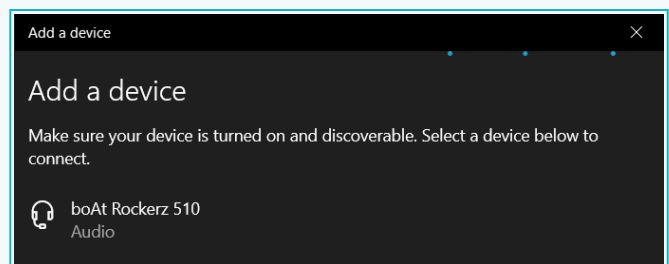
3 Switch the Bluetooth on for the device to be connected. For example, a mobile phone, headphones, or a speaker.

Activity 2

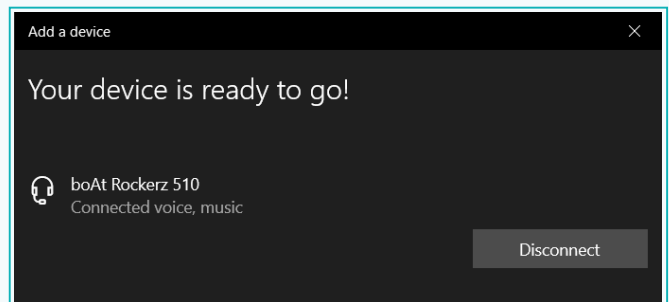
4 Click on 'Add a Device'.



5 Click on the name of the device to be added.



6 The device should now be successfully connected!



It is very easy to transfer data and information to any device in the world through the internet. Many kinds of internet connections are used in different places all around us.

Healthy Practices:

- Switch off the Bluetooth when you are not using it.
- Do not pair your device with unknown devices.

2. Network Connections

I didn't know there were so many ways in which we can share information over the internet.

Yes, Nishit, that's the best part about the internet.



Tech Flash

- **Internet** : The internet is a global network of computers connected to each other over a set of standard rules.
- **Computer Network** : A group of inter-connected computers is called a computer network.
- **Port** : A port is a point where a cable connects to a device.
- **Ethernet** : Ethernet is a standard cable that is used to connect to the internet.
- **Router** : A router is a device which connects more than one device to internet. It also acts as connection between two networks.
- **Modem** : A modem is a device used to connect to the internet.
- **LAN** : Local Area Network is a small area network where computers are connected to each other.
- **WAN** : Wide Area Network is a large area network where computers on different LANs are connected to each other.
- **PAN** : Personal Area Network is a network between devices that are used by a single person.
- **Bluetooth** : Bluetooth is a technology that wirelessly shares files between connected devices.
- **WiFi** : WiFi or Wireless Fidelity is a method that connects to the internet wirelessly (without cables).